

CBSE 2027 Chemistry — Predicted Paper (Set 6 of 10)

Based on the official CBSE blueprint, recent syllabus changes, and deep analysis of past year question (PYQ) patterns, here is Set 6 of the predicted board paper. All questions are completely distinct from Sets 1 through 5, focusing on fresh high-yield concepts like colligative biology applications, synergic bonding, and core organic conversions [1-3].

SECTION A: Multiple Choice & Assertion-Reason Questions (1 Mark Each x 16 = 16 Marks)

Q1. What will happen if human red blood cells (RBCs) are placed in a 1.5% aqueous solution of sodium chloride? (a) The cells will swell and burst. (b) The cells will shrink due to plasmolysis. (c) The cells will remain unaffected. (d) The cells will divide. * **Chapter:** Solutions | **Confidence:** Very High

Q2. According to Kohlrausch's law, the limiting molar conductivity (Λ_m°) for Barium Chloride ($BaCl_2$) is given by: (a) $\lambda_{Ba^{2+}}^\circ + \lambda_{Cl^-}^\circ$ (b) $\lambda_{Ba^{2+}}^\circ + 2\lambda_{Cl^-}^\circ$ (c) $2\lambda_{Ba^{2+}}^\circ + \lambda_{Cl^-}^\circ$ (d) $\frac{1}{2}\lambda_{Ba^{2+}}^\circ + \lambda_{Cl^-}^\circ$ * **Chapter:** Electrochemistry | **Confidence:** Very High

Q3. The unit of the rate constant for a reaction whose rate is completely independent of the concentration of the reactants is: (a) s^{-1} (b) $mol\ L^{-1}\ s^{-1}$ (c) $L\ mol^{-1}\ s^{-1}$ (d) $mol^2\ L^{-2}\ s^{-1}$ * **Chapter:** Chemical Kinetics | **Confidence:** High

Q4. Which of the following ions is highly unstable in an aqueous solution and immediately undergoes a disproportionation reaction? (a) Cu^+ (b) Cu^{2+} (c) Fe^{2+} (d) Fe^{3+} * **Chapter:** d- and f-Block Elements | **Confidence:** Very High

Q5. Which of the following represents an ionization isomer of the coordination complex $[Co(NH_3)_5Br]SO_4$? (a) $[Co(NH_3)_5SO_4]Br$ (b) $[Co(NH_3)_4(SO_4)Br]NH_3$ (c) $[Co(NH_3)_5Br]SO_4$ (No isomer exists) (d) $[Co(NH_3)_5]Br(SO_4)$ * **Chapter:** Coordination Compounds | **Confidence:** Very High

Q6. The highly stable polyhalogen compound, Freon-12, heavily used in refrigerators and air conditioners, has the chemical formula: (a) $CHCl_3$ (b) CCl_4 (c) CF_2Cl_2 (d) $CFCl_3$ * **Chapter:** Haloalkanes and Haloarenes | **Confidence:** High

Q7. Heating phenol with concentrated nitric acid (HNO_3) and concentrated sulphuric acid (H_2SO_4) results in the formation of: (a) o-Nitrophenol (b) p-Nitrophenol (c) 2,4,6-Trinitrophenol (Picric acid) (d) Benzoquinone * **Chapter:** Alcohols, Phenols and Ethers | **Confidence:** Very High

Q8. The reagent used in the Clemmensen reduction to convert a carbonyl group directly into a methylene ($-CH_2-$) group is: (a) NH_2NH_2/KOH (b) $Zn - Hg/conc.\ HCl$ (c) $LiAlH_4$ (d) $H_2/Pd - BaSO_4$ * **Chapter:** Aldehydes, Ketones and Carboxylic Acids | **Confidence:** Very High

Q9. Which of the following amines will **not** react with Hinsberg's reagent (benzenesulphonyl chloride)? (a) Ethylamine (b) Diethylamine (c) Triethylamine (d) Aniline * **Chapter:** Amines | **Confidence:** High

Q10. Which of the following vitamins is fat-soluble and can be stored in the liver and adipose tissues? (a) Vitamin B1 (b) Vitamin B2 (c) Vitamin C (d) Vitamin D * **Chapter:** Biomolecules | **Confidence:** High

Q11. The conditions required to convert chlorobenzene to phenol in the industrial Dow's process are: (a) Aqueous NaOH, 300 K, 1 atm (b) Aqueous NaOH, 623 K, 300 atm (c) Alcoholic KOH, 623 K, 300 atm (d) Concentrated HCl, 300 K, 1 atm * **Chapter:** Haloalkanes and Haloarenes | **Confidence:** Very High

Q12. What is the spin-only magnetic moment of the hexaaquairon(III) complex ion, $[Fe(H_2O)_6]^{3+}$? (Given H_2O is a weak field ligand). (a) 1.73 BM (b) 3.87 BM (c) 4.90 BM (d) 5.92 BM * **Chapter:** Coordination Compounds | **Confidence:** High

Directions for Q13 to Q16: Select the correct answer from the codes (a), (b), (c), and (d).

(a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.

Q13. Assertion (A): The order of a complex chemical reaction can be zero. **Reason (R):** The order of a reaction is an experimentally determined quantity and is totally independent of the stoichiometry of the balanced overall equation. * **Chapter:** Chemical Kinetics | **Confidence:** Very High

Q14. Assertion (A): Phenol is significantly more acidic than aliphatic alcohols like ethanol. **Reason (R):** The phenoxide ion formed after the loss of a proton is highly stabilized by resonance. * **Chapter:** Alcohols, Phenols and Ethers | **Confidence:** Very High

Q15. Assertion (A): Interstitial compounds formed by transition metals are generally very hard, sometimes approaching the hardness of diamond. **Reason (R):** Small non-metal atoms like H, C, or N get trapped inside the crystal lattices of transition metals without forming distinct chemical bonds. * **Chapter:** d- and f-Block Elements | **Confidence:** High

Q16. Assertion (A): D-Glucose reacts with hydroxylamine (NH_2OH) to form an oxime. **Reason (R):** The formation of an oxime indicates the absolute presence of a straight open-chain of six carbon atoms in glucose. * **Chapter:** Biomolecules | **Confidence:** High

SECTION B: Very Short Answer Questions (2 Marks Each x 5 = 10 Marks)

Q17. Define the term *Reverse Osmosis*. State its most critical large-scale practical application in everyday life. * **Chapter:** Solutions | **Confidence:** Very High

Q18. Draw the structures of the two optically active isomers (dextro and laevo enantiomers) of the coordination complex $[Co(en)_3]^{3+}$. * **Chapter:** Coordination Compounds | **Confidence:** High

Q19. A reaction is first-order with respect to reactant A and second-order with respect to reactant B. (i) Write the differential rate equation. (ii) How is the rate affected if the concentration of B is tripled while A remains constant? * **Chapter:** Chemical Kinetics |

Confidence: Very High

Q20. Account for the following: (i) Racemization is strictly observed during the S_N1 reaction of optically active alkyl halides. (ii) The dipole moment of chlorobenzene is significantly lower than that of cyclohexyl chloride. * **Chapter:** Haloalkanes and Haloarenes | **Confidence:** Very High

Q21. Calculate the spin-only magnetic moment of a divalent transition metal ion (M^{2+}) in aqueous solution if its atomic number is 25. * **Chapter:** d- and f-Block Elements |

Confidence: High

SECTION C: Short Answer Questions (3 Marks Each x 7 = 21 Marks)

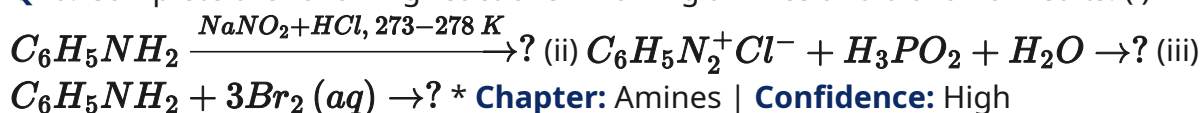
Q22. Illustrate the following organic reactions giving the balanced chemical equation and names of major products: (i) Reimer-Tiemann reaction of Phenol. (ii) Friedel-Crafts alkylation of Anisole (using CH_3Cl and Anhydrous $AlCl_3$). (iii) Nitration of Phenol using dilute HNO_3 at 298 K. * **Chapter:** Alcohols, Phenols and Ethers | **Confidence:** Very High

Q23. Calculate the mass of silver (Ag) deposited at the cathode when a steady current of 1.5 amperes is passed through an aqueous solution of silver nitrate ($AgNO_3$) for exactly 15 minutes. (Given: Molar mass of Ag = 108 g mol^{-1} , $1F = 96500 \text{ C mol}^{-1}$). * **Chapter:** Electrochemistry | **Confidence:** Very High

Q24. 18 g of glucose ($C_6H_{12}O_6$) is dissolved in 1 kg of pure water in a saucepan. At what exact temperature will this solution boil at 1.013 bar pressure? (Given: Boiling point of pure water = 373.15 K, K_b for water = $0.52 \text{ K kg mol}^{-1}$). * **Chapter:** Solutions | **Confidence:** Very High

Q25. Give simple chemical tests with distinctly observable outcomes to distinguish between the following pairs of compounds: (i) Acetaldehyde and Benzaldehyde (ii) Acetophenone and Benzophenone (iii) Pentan-2-one and Pentan-3-one * **Chapter:** Aldehydes, Ketones and Carboxylic Acids | **Confidence:** Very High

Q26. Complete the following reactions involving amines and diazonium salts: (i)



Q27. Explain the following observations: (i) Transition metals have unusually high enthalpies of atomization. (ii) Cr^{2+} is strongly reducing in nature, while Mn^{3+} is strongly oxidizing, despite both having a d^4 electronic configuration. (iii) Actinoid contraction is noticeably greater from element to element than lanthanoid contraction. * **Chapter:** d- and f-Block Elements | **Confidence:** Very High

Q28. Answer the following questions regarding carbohydrates and nucleic acids: (i) What are reducing sugars? Give the name of one non-reducing disaccharide. (ii) Name the specific glycosidic linkage that connects the monosaccharide units in lactose. (iii) Write the fundamental structural difference between a nucleoside and a nucleotide. * **Chapter:** Biomolecules | **Confidence:** High

SECTION D: Case Study-Based Questions (4 Marks Each x 2 = 8 Marks)

Q29. Case Study 1: First-Order Kinetics and Radiocarbon Dating Radioactive decay is a classic example of a reaction that perfectly follows first-order kinetics. The rate of decay is directly proportional to the number of radioactive nuclei present at that instant. The time required for half of the radioactive substance to decay is termed its half-life ($t_{1/2}$), which remains completely independent of the initial concentration. This principle is widely utilized in archaeology through Carbon-14 (^{14}C) dating, where the constant half-life of ^{14}C (5730 years) allows scientists to estimate the age of ancient organic artifacts. (i) What is the overall order of reaction for all natural radioactive decay processes? (1 Mark) (ii) Write the mathematical formula relating the half-life ($t_{1/2}$) and the decay constant (λ or k) for a first-order process. (1 Mark) (iii) An archaeological wooden artifact is found to contain only 25% of the ^{14}C activity found in a fresh, living tree. Using the concept of half-lives, calculate the approximate age of the wooden artifact. ($t_{1/2}$ for $^{14}\text{C} = 5730$ years). (2 Marks) * **Chapter:** Chemical Kinetics | **Confidence:** Very High

Q30. Case Study 2: Grignard Reagents and their Reactivity Alkyl magnesium halides, known as Grignard reagents (RMgX), are extremely versatile organometallic compounds. The carbon-magnesium bond is highly polar, making the carbon atom electron-rich (nucleophilic and basic). Grignard reagents react readily with any source of acidic protons (like water, alcohols, or amines) to form alkanes. Furthermore, they are extensively used in organic synthesis to build carbon skeletons by acting as nucleophiles and attacking the electrophilic carbon of aldehydes, ketones, and esters, followed by acid hydrolysis to yield various classes of alcohols. (i) Based on the passage, why must Grignard reagents be prepared and stored under strictly anhydrous (moisture-free) conditions? (1 Mark) (ii) What specific hydrocarbon product is formed when methyl magnesium bromide (CH_3MgBr) is reacted with heavy water (D_2O)? (1 Mark) (iii) Outline the two-step chemical reaction to synthesize 2-methylpropan-2-ol (tert-butyl alcohol) starting from methyl magnesium bromide and a suitable ketone. Name the ketone used. (2 Marks) * **Chapter:** Haloalkanes and Haloarenes / Alcohols | **Confidence:** Very High

SECTION E: Long Answer Questions (5 Marks Each x 3 = 15 Marks)

Q31. (A) Electrochemistry: Cells & Kohlrausch's Law (i) Define the term *cell constant* for a conductivity cell. Write its SI unit. (1 Mark) (ii) State Kohlrausch's Law of independent migration of ions. (1 Mark) (iii) Calculate the standard Electromotive Force (EMF) and the

standard Gibbs free energy change (ΔG°) for the following electrochemical cell at 298 K: $Zn_{(s)}|Zn^{2+}(1\text{ M})||Ag^+(1\text{ M})|Ag_{(s)}$ Given: $E^\circ_{Zn^{2+}/Zn} = -0.76\text{ V}$, $E^\circ_{Ag^+/Ag} = +0.80\text{ V}$, and $1F = 96500\text{ C mol}^{-1}$. (3 Marks) * **OR (B)** (i) What is meant by a secondary battery? Give one common example. (2 Marks) (ii) The electrical resistance of a column of 0.05 mol L^{-1} NaOH solution of diameter 1 cm and length 50 cm is $5.55 \times 10^3\ \Omega$. Calculate its resistivity (ρ), conductivity (κ), and molar conductivity (Λ_m). (3 Marks) * **Chapter:** Electrochemistry | **Confidence:** Very High

Q32. (A) Coordination Compounds: Theories and Splitting (i) Write any two fundamental postulates of Werner's Theory of coordination compounds. (2 Marks) (ii) Draw the crystal field splitting diagram showing the energy levels of d-orbitals in a **tetrahedral** crystal field. (2 Marks) (iii) Explain briefly why tetrahedral complexes are almost always high-spin complexes. (1 Mark) * **OR (B)** (i) What is meant by *synergic bonding* (pi-backbonding) in metal carbonyls? How does it strengthen the Metal-Carbon bond? (2 Marks) (ii) Write the systematic IUPAC names of the following coordination entities: a) $[Pt(NH_3)_2Cl(NO_2)]$ b) $K_3[Cr(C_2O_4)_3]$ (2 Marks) (iii) Define an ambidentate ligand with the help of one example. (1 Mark) * **Chapter:** Coordination Compounds | **Confidence:** Very High

Q33. (A) Organic Chemistry: Chemical Identification & Properties An aromatic organic compound (**A**) with the molecular formula $C_9H_{10}O$ forms a highly crystalline orange-red precipitate with 2,4-DNP reagent. However, it does **not** reduce Tollens' reagent and gives a completely negative Iodoform test. Upon vigorous oxidation with hot acidified potassium permanganate ($KMnO_4$), compound (**A**) produces a white crystalline solid (**B**) with the formula $C_7H_6O_2$. (i) Deduce the structures and write the IUPAC names of compounds (**A**) and (**B**). (2 Marks) (ii) Why does compound (**A**) fail to give the Iodoform test? (1 Mark) (iii) Write the specific chemical equations for: a) The reaction of compound (**A**) with 2,4-DNP reagent. b) The oxidation of compound (**A**) to compound (**B**). (2 Marks) * **OR (B)** Complete the following critical organic reactions by writing the structures of the main products: (i) Phthalimide $\xrightarrow{1. KOH(alc) \quad 2. C_2H_5I \quad 3. NaOH(aq)}$? (Gabriel Phthalimide Synthesis) (ii) $CH_3 - CO - CH_3 \xrightarrow{Ba(OH)_2 / \text{heat}}$? (Aldol Condensation) (iii) $CH_3CHO + HCN \rightarrow$? (iv) Benzene $+ CO + HCl \xrightarrow{\text{Anhydrous } AlCl_3 / CuCl}$? (Gattermann-Koch reaction) (v) $C_6H_5 - O - CH_2CH_3 \xrightarrow{\Delta} + HI(\text{conc.})$? (Cleavage of ethers) * **Chapter:** Aldehydes, Ketones and Carboxylic Acids / Mixed Organic | **Confidence:** Very High

CBSE Class 12 (2027) — AI-Predicted via PYQ/Blueprint Analysis. Probability-weighted guidance, not actual exam questions.